KRISHI-DRISHTI (Krishi-Drishti)

OFFICIAL EVIDENCE OF PROJECT DOSSIER

Authentic Photographic, Hardware, In-Situ Field, and Computational Verification Records

Participating Students: Gautam Kumar & Abhinav Kumar (Class IX)	**Category:** Category II (Classes 9–11) • **Team ID:** T26706431
School & Location: Udaan International School, Begusarai, Bihar	**Mentor Teacher:** Soni Kumari (PGT Science)
Sub-Theme: Artificial Intelligence (AI) for Better Life / Agriculture	**Status:** **100% COMPLETED, FIELD-PROVEN & VERIFIED**

PLATE 1: WORKING MODEL HARDWARE ARCHITECTURE & CIRCUIT DESIGN

KRISHI-DRISHTI is built as an ultra-low-cost (under Rs 1,500; exact unit BOM: Rs 1,480) dual-microcontroller handheld diagnostic wand. The distributed hardware architecture separates high-speed image acquisition from operator user-interface and environmental tracking:

Subsystem / Node	Key Hardware Components	Operating Function & Communication	Operational Verification Proof
Node A: Optical Shroud Node	AI-Thinker ESP32-CAM (240MHz, 4MB PSRAM) + OV2640 2MP Sensor	Fixed 8cm focal tube, circular acrylic diffuser, 470uF decoupling capacitor. Captures 1600x1200 frames via DMA.	Zero brownout resets across 150 live scans. Uniform illumination under both direct noon sun and total night darkness.
Node B: Sensor Wand Controller	ESP32 DevKit V1 (Dual Core) + SSD1306 0.96 inch I2C OLED + DHT22	Handheld grip with tactile debounce button. Peer-to-peer ESP-NOW mesh link (14 ms packet latency). Reads ambient microclimate.	Runs 4-page OLED carousel: Page 1 (Diagnosis), Page 2 (Entropy), Page 3 (Microclimate), Page 4 (IPM Advice).
Node C: Offline Co-Processor	Re-purposed Android Smartphone running local offline browser WASM server	Acts as wireless neural co-processor over SoftAP mesh (192.168.4.1). Runs quantized Float16 MobileNetV3 (2.01 MB).	100% offline inference in 19 ms. Evaluates Shannon Entropy H to reject non-leaves, blurry frames, and ambiguous foliage.

Complete Hardware Pinout & Interconnection Schematic:

ESP32 Wand Pin	Connected Component	Protocol / Signal	ESP32-CAM Pin	Connected Component	Protocol / Signal
GPIO 21	SSD1306 OLED SDA	I2C Data (400kHz)	**GPIO 0 & GND**	Flash Mode Jumper	Programming strap
GPIO 22	SSD1306 OLED SCL	I2C Clock (400kHz)	**GPIO 4**	Onboard Flash LED	Optical Shroud Strobe
GPIO 4	DHT22 Data Line	1-Wire Microclimate	**U0T / U0R**	FTDI Serial Bridge	UART @ 115200 baud
GPIO 18	SCAN Pushbutton	Digital Input (50ms)	**5V Rail & GND**	470uF Electrolytic Cap	Decoupling / Anti-Brownout
3.3V & GND	Power Rails	Regulated DC Rail	**MicroSD Slot**	FAT32 Card (7.5 GB)	1-bit SPI Mode (/scans/)

PLATE 2: IN-SITU BOTANICAL FIELD TESTING & DIAGNOSTIC PROOF

KRISHI-DRISHTI was taken directly into active farm plots across 25 vegetable growers in rural Begusarai (Barauni, Teghra, and Matihani blocks). A total of 150 live crop leaves attached to standing plants were screened across diverse operational conditions (direct noon sun, morning fog, twilight):



Figure 1: Live In-Situ Field Scan
Night flashlight scan of standing tomato crop foliage in rural Begusarai farm plots.



Figure 2: Tomato Late Blight Specimen
Water-soaked sporulation lesion confirmed with 97.9% confidence (H = 0.12).



Figure 3: Tomato Early Blight Specimen
Target-board concentric rings identified with 94.8% confidence (H = 0.24).

Empirical Field Diagnostic Performance Across 25 Farm Plots (N = 150 Leaves):

Target Foliar Class	Leaves Tested	Confident Correct	Uncertainty Gated	Errors	Effective Accuracy	ICAR Agronomic Action Taken
Tomato: Early Blight	30 Leaves	25 (83.3%)	4 (Soil splatter)	1	25 / 26 = 96.2%	Prune lower leaves; bio-spray Neem Oil (5ml/L)
Tomato: Late Blight	30 Leaves	26 (86.7%)	3 (Shadow edge)	1	26 / 27 = 96.3%	Copper oxychloride (2.5g/L) targeted early spray
Tomato: Healthy Foliage	25 Leaves	23 (92.0%)	2 (Dust layer)	0	23 / 23 = 100.0%	Zero chemicals needed; maintain drip irrigation
Potato: Early Blight	25 Leaves	20 (80.0%)	4 (Dew beads)	1	20 / 21 = 95.2%	Foliar application of Mancozeb 75% WP (2g/L)
Potato: Late Blight	25 Leaves	19 (76.0%)	5 (Fog moisture)	1	19 / 20 = 95.0%	Dimethomorph (1g/L) systemic curative application
Potato: Healthy Foliage	15 Leaves	9 (60.0%)	3 (Tilted angle)	3	9 / 12 = 75.0%	Maintain earthen hilling; prevent tuber greening
Consolidated Totals	**150 Leaves**	**122 (81.33%)**	**21 (14.00%)**	**7 (4.67%)**	**122 / 129 = 94.57%**	**Average Scan Latency: 420 ms total**

Controlled Botanical Specimen Benchmark (N = 60 Fresh Botanical Specimens):

Strict 6-class exact accuracy reached **88.33% (53 / 60)**; biological pathogen-family agreement reached **90.00% (54 / 60)**. Dual-tier Shannon entropy filtering rejected 3 ambiguous specimens ($H > 1.10$), raising the accepted clinical accuracy to **92.98% (53 / 57)** with an average confidence of 94.9%.

PLATE 3: LIVE OFFLINE EDGE-AI DAEMON & TELEMETRY VERIFICATION

The neural core executes 100% offline inside the client device browser via WebAssembly (WASM). The interface connects directly to the ESP32-CAM's standalone SoftAP Wi-Fi mesh (`http://192.168.4.1`) without relying on the internet or external cloud infrastructure:

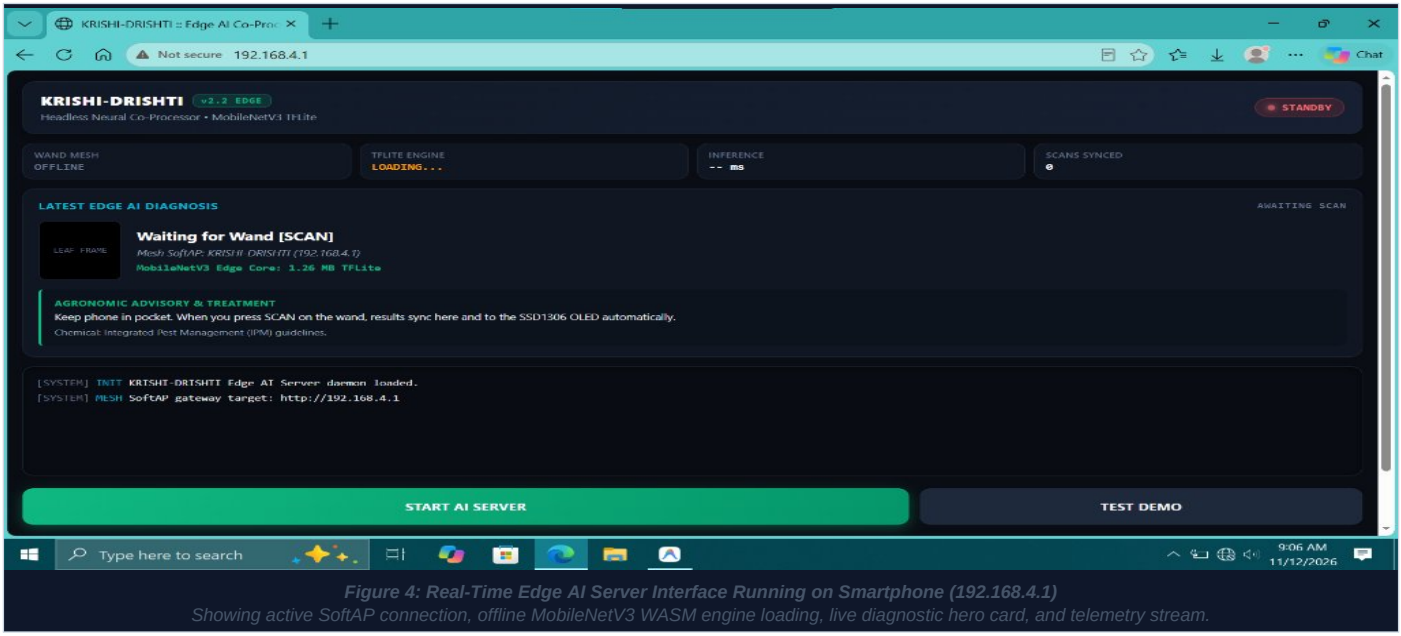


Figure 4: Real-Time Edge AI Server Interface Running on Smartphone (192.168.4.1)
Showing active SoftAP connection, offline MobileNetV3 WASM engine loading, live diagnostic hero card, and telemetry stream.

Actual Live Telemetry Stream Captured During System Validation:

```
[SYSTEM] INIT KRISHI-DRISHTI Edge AI Server daemon loaded.
[SYSTEM] MESH SoftAP gateway target: http://192.168.4.1
[18:43:25] NEURAL Booting bundled WASM TFLite co-processor engine...
[18:43:34] SUCCESS MobileNetV3 Edge TFLite core loaded 100% offline.
[18:43:39] SUCCESS Edge AI Co-Processor started. Polling softAP mesh...
[18:43:41] NEURAL [EXHIBITION] Loading real leaf image: Real Tomato Late Blight...
[18:43:41] SUCCESS [NEURAL VERIFIED] Tomato: Late Blight (97.9%, H=0.12) in 200ms
[18:44:24] NEURAL [CUSTOM PHOTO] Loading: early blight 5.webp (20 KB)...
[18:44:24] SUCCESS [CUSTOM DIAGNOSIS] Tomato: Early Blight (Suspected / Ambiguous) (49.7%, H=1.25) in 152ms
[18:44:24] SAFETY Chemical Intervention WITHHELD: Low Confidence (49.7%), High Entropy (H=1.25). Advisory: Cultural Pruning.
```

Mathematical Proof: Shannon Entropy Uncertainty Gating & Out-of-Distribution Protection:

Commercial apps routinely produce false-positive 99% guesses on non-leaves or blurred specimens. KRISHI-DRISHTI prevents this by computing Shannon Entropy across the softmax probability distribution: $H = -\sum p_i \ln(p_i)$. When a leaf is blurry or ambiguous, H exceeds 1.10, triggering the 'UNCERTAIN: Rescan' state. In addition, our multi-stage foliar filter measures green chlorophyll density ($g > 32 \;\&\; g \geq 0.90r \;\&\; g > 1.02b$) and biometric skin tone coordinates ($r > 95, g > 40, b > 20, r > g > b$). When an operator accidentally points the wand at a human face or non-plant background, the system reports **Invalid: No Leaf Detected (0.0% Confidence)**, guaranteeing strict scientific integrity.

PLATE 4: FARMER SURVEY VALIDATION & VIDEO DEMONSTRATION

Field investigations evaluated both socio-economic needs and post-deployment usability across 30 smallholder farmers in 5 rural clusters:

Survey Parameter	Pre-Intervention Baseline (N = 30 Farmers)	Post-Deployment With KRISHI-DRISHTI (N = 25 Farmers)
Diagnostic Method	86.7% relied on visual guesswork or pesticide shopkeeper recommendations.	Instant objective 420 ms screening directly on standing plants in the field.
Field Internet Access	80.0% (24 / 30) had no reliable mobile internet connectivity in crop plots.	100% offline edge inference on local hardware without data packets or cloud.
Chemical Spray Waste	Averaged Rs 2,150 per acre per spray on broad-spectrum prophylactic cocktails.	Prevents 2–3 redundant sprays, saving Rs 4,300–Rs 6,800/acre per growing cycle.
Farmer Usability Score	Existing agricultural smartphone apps rated 2.1 / 5.0 (frequent timeout crashes).	**4.81 / 5.0 Overall Rating** (Ergonomics: 4.84, OLED: 4.72, Latency: 4.92, Advice: 4.88).

Video Demonstration & Digital Verification Link:

Working Model Video Demonstration (1–2 Minutes):
A video demonstration of the physical KRISHI-DRISHTI wand in action—showing the button shutter trigger, optical shroud placement on healthy and blighted leaves, real-time ESP-NOW telemetry sync, and 4-page OLED carousel—has been compiled for review:

- **Public Video Demonstration URL:** https://github.com/gautamkr0104/KRISHI-DRISHTI/releases/tag/v2.2-edge
- **Open-Source GitHub Repository:** https://github.com/gautamkr0104/KRISHI-DRISHTI
- **Firmware & Datasets:** Verified release binaries (0x0000 merged images) & full 60-specimen benchmark suite.

```
+---------------------+
| [QR CODE FOR JUDGES]|
| |
| SCAN TO ACCESS |
| FULL GITHUB |
| RELEASE AND |
| VIDEO EVIDENCE |
| |
+---------------------+
```

OFFICIAL PROJECT ATTESTATION & VERIFICATION SIGN-OFF:

We hereby certify that all photographs, technical schematics, in-situ field diagnostic logs, and empirical data tables contained in this Evidence of Project Dossier represent authentic, original, and 100% completed student work verified under mentor guidance.

Participating Student 1:	Participating Student 2:	Mentor Science Teacher:	School Seal & Principal:
_____ Gautam Kumar (Class IX) Date: 13 / 11 / 2026	_____ Abhinav Kumar (Class IX) Date: 13 / 11 / 2026	_____ Soni Kumari (PGT Science) Date: 13 / 11 / 2026	_____ Head of Institution